# Patrick Thangarajah

Product Design | Prototyping | Testing & Validation | Simulation

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**Hiring Team** *11 November 2026*

Planar Motor Inc.
Richmond, BC

**Mechanical Design Engineer/EIT/Specialist**

Dear Planar Motor Hiring Team,

I recently completed an MASc in Mechanical and Mechatronics Engineering at the University of Waterloo, and the work I enjoy most is designing hardware, building it, and proving on the bench that it does what it should. Planar Motor's levitating movers sit right where mechanical design meets motion control, and the range of industries you serve, from battery production to medical devices, is the kind of applied engineering I want to grow in.

Building test apparatus is where most of my hands-on experience comes from. For my UBC capstone, I prototyped silicone and agar bladder phantoms and built the control and data acquisition interface for a pressurized test system, which ran stable, repeatable fill and void cycles up to 4.75 kPa. At Echoscope I carried that platform forward, designing its PCB and writing five manufacturing and verification procedures that the client approved and used. On the UBC drone team, I designed a payload drop system in SolidWorks and Onshape, iterated it through 3D printed builds, and cut its weight by 50% by repurposing components from earlier designs.

I also understand production environments. As a Mechanical Engineering Intern at GN Corporation, I wrote work instructions and SOPs from engineering drawings and customer specifications, and reduced fiberglass composite cutting waste by 50% through process analysis. My graduate research added the problem-solving side: I validated an LS-DYNA model against physical sled tests, traced model-to-test discrepancies back to positioning, restraint fit and response timing, and wrote LS-PrePost scripts that cut a repeated setup from 7 steps to 4. I work comfortably in SolidWorks and with 3D printing, a lathe and a mill, and I am keen to build my CNC machining and welding experience.

I am eligible to work in Canada, available to start immediately, and happy to relocate to Richmond. My resume and transcripts are attached, and I would welcome the chance to talk about how I can contribute to your design team.

Sincerely,

**Patrick Thangarajah**